EDC · T7

Multistage amplifiers

Cheat sheet · Day 14 · Fri 6 Nov · pack E8

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
MULTISTAGE (T7)
Cascade: AvT = Av1*Av2*...  each Av computed with the NEXT stage's Zi as load
  Avs = Zi1/(Zi1+Rs) * AvT      AvL = AvNL * RL/(RL+Zo)    (two-port; Zo(CE)=RC)
  Two-port chain: Avs = [Zi1/(Zi1+Rs)] AvNL1 [Zi2/(Zi2+Zo1)] AvNL2 [RL/(RL+Zo2)]
Zi:  CE bypassed   R1||R2||b*re          CE unbypassed  R1||R2||b(re+RE)
     EF            RB||b(re + RE||RL)
Av:  CE bypassed  -(RC||Rload)/re        CE unbypassed  -(RC||Rload)/(re+RE) ~ -(RC|
     EF   (RE||RL)/(re+RE||RL) ~ 1       CB  +RC/re
Phase: CE inverts; CE-CE = +ve gain; EF, CB no inversion
CHECK: |Vo| = |AvT| Vi < available swing (~VCE)? else clipping
Lecture Wk5 ex (both bypassed, b=200, re=6.34): Av1=-104, Av2=-347, AvT=36.1e3; RL=4
Prac4 Fig1 (stage 2 UNBYPASSED): Zi2=3.52k, Av1=-212, Av2=-2.19 -> 463; RL=4.7k -> 31
Darlington: bD=b1*b2, VBE=2 drops (1.4 V; 1.6 V Boylestad)
  IB=(VCC-VBE)/(RB+bD*RE), Zi=RB||bD*RE, Ai~bD*RB/(RB+bD*RE), Av~1
Feedback pair: IB1=(VCC-VEB1)/(RB+b1*b2*RC), one VBE drop
Cascode: VB1=R3/(R1+R2+R3)*VCC, VB2=(R2+R3)/(R1+R2+R3)*VCC, IE=(VB1-0.7)/RE same both
  VC1=VB2-0.7, Av1=-1, Av2=+RC/re, Av~-RC/re; benefit = bandwidth (low Miller), not g
Direct-coupled: VB2 = VC1 = VCC - IC1*RC1 (check IB2 << IC1, else KCL at node)
Mirror: IREF=(VCC - VB1)/R, VB1 = emitter rail + 0.7;  Io = IREF/(1+2/b) -> IREF
  Design: R = (VCC - VB1)/[Io(1+2/b)]; keep output BJT active (VCE>=0.2)
npn: VE = VB - 0.7     pnp: VE = VB + 0.7     beta = inf -> IB = 0
```